ENGINEERING ECONOMICS

CE 655 | Year III / Part II

Chapter 8: Inflation and Its Effects on Engineering Economy

Questions & Answers — TU Past Exam Papers (2070–2082)

Tribhuvan University | Institute of Engineering

Chapter 8 Syllabus Overview (4 Hours)

Topics:

a) Introduction to Inflation — Definition, Causes, Measurement (CPI, Inflation Rate)

b) Purchasing Power and Its Relationship with Inflation

c) Constant-Worth (Real) vs. Actual (Nominal/Then-Current) Dollars

d) Real vs. Nominal Interest Rates — Relationship and Conversion

e) Economic Analysis in Constant-Worth Dollars

f) Economic Analysis in Actual Dollars (Nominal Dollar Analysis)

g) Effect of Inflation on After-Tax Analysis

h) Capital Recovery with Inflation

Section A: Inflation — Concepts and Terminology

Q1. Define inflation. What are its main causes? How is inflation measured using the Consumer Price Index (CPI) and the inflation rate formula? [2082 Kartik Q11a, 2081 Ashwin Q11a, 2079 Jestha Q11a — 4 Marks]

Answer:

Definition of Inflation

Inflation is the sustained, general rise in the price level of goods and services in an economy over time, resulting in a decrease in the purchasing power of money. In other words, a unit of currency buys fewer goods and services today than it did in the past. For engineering economics, inflation is important because project cash flows spread over many years will have different purchasing powers depending on when they occur, and ignoring inflation can lead to incorrect economic decisions.

Causes of Inflation

1. Demand-Pull Inflation: When aggregate demand in an economy exceeds aggregate supply, prices are pulled upward. This often occurs during periods of rapid economic growth, increased government spending, or credit expansion.
2. Cost-Push Inflation: Rising production costs (wages, raw materials, energy) force producers to raise prices to maintain profit margins. Oil price shocks are a classic example.
3. Built-In (Wage-Price) Inflation: Workers demand higher wages to compensate for rising living costs, which increases production costs, which leads to higher prices — a self-perpetuating cycle.
4. Monetary Inflation: When the money supply grows faster than the real economy (excessive money printing), more money chases the same quantity of goods, driving prices up.
5. Imported Inflation: Rising prices of imported goods and raw materials push up domestic prices, especially in import-dependent economies.
6. Supply Disruptions: Natural disasters, wars, pandemics, or supply chain disruptions reduce the availability of goods, pushing prices upward.

Measuring Inflation — Consumer Price Index (CPI)

The Consumer Price Index (CPI) measures the average change in the price of a fixed 'basket' of representative goods and services purchased by typical households, relative to a chosen base year. It is the most widely used measure of inflation.

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CPI_n = (Cost of basket in year n / Cost of same basket in base year) × 100
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A CPI of 150 means prices are 50% higher than in the base year (CPI = 100).

Inflation Rate Formula

The general inflation rate (f) for a given period is the percentage change in CPI from one period to the next:

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f_n = (CPI_n - CPI_{n-1}) / CPI_{n-1}  × 100%
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Where: f_n = inflation rate in year n, CPI_n = Consumer Price Index in year n, CPI_{n-1} = Consumer Price Index in year (n−1).

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Average annual inflation rate over N years:   (1 + f_avg)^N =
                                CPI_N / CPI_0
                f_avg = (CPI_N / CPI_0)^(1/N) - 1
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Example: CPI rose from 150 to 165 in one year. Inflation rate f = (165−150)/150 = 0.10 = 10% per year.

Q2. Define purchasing power. How does inflation affect purchasing power? Explain with a numerical example. [2081 Chaitra Q11a, 2079 Ashwin Q11a, 2077 Chaitra Q11a — 3 Marks]

Answer:

Purchasing Power — Definition

Purchasing power is the quantity of goods and services that a unit of currency can buy at a given point in time. Purchasing power and the price level are inversely related — as the general price level rises (inflation), the purchasing power of money falls.

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Purchasing Power of Rs. 1  =  1 / Price Index  (relative to base
                              year)
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In practical terms, if Rs. 1,000 could buy a certain basket of goods in year 0, the same basket will cost more than Rs. 1,000 in future years due to inflation. The Rs. 1,000 in year n has less real purchasing power than Rs. 1,000 in year 0.

Effect of Inflation on Purchasing Power

If the annual inflation rate is f, then the purchasing power of Rs. 1 received n years from now, expressed in terms of today's purchasing power, is:

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Purchasing Power (today's Rs.) = 1 / (1+f)^n
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Numerical Example

Suppose annual inflation rate f = 8% per year. Find the purchasing power of Rs. 50,000 received 5 years from now, in terms of today's rupees.

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Purchasing Power = 50,000 / (1+0.08)^5 = 50,000 / 1.4693 = Rs. 34,029
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Interpretation: Rs. 50,000 received 5 years from now has the same purchasing power as only Rs. 34,029 today. Inflation has eroded Rs. 15,971 of real value.

Alternatively: If you invest Rs. 34,029 today at 8% for 5 years, you get Rs. 50,000 — but this Rs. 50,000 buys the same goods as Rs. 34,029 does today. You have not gained purchasing power; you have merely kept pace with inflation.

Year	Rs. in Hand	Purchasing Power (in Year-0 Rs.)	Loss of Real Value
0	50,000	50,000	—
1	50,000	46,296	3,704
3	50,000	39,692	10,308
5	50,000	34,029	15,971
10	50,000	23,160	26,840

Q3. Distinguish between actual dollars (nominal/then-current dollars) and constant-worth dollars (real dollars). Why is this distinction important in engineering economic analysis? [2082 Kartik Q11b, 2080 Chaitra Q11a, 2078 Chaitra Q11a — 4 Marks]

Answer:

Actual Dollars (Nominal / Then-Current Dollars)

Actual dollars (also called nominal dollars or then-current dollars) represent the cash amounts that are actually paid or received at the time of each transaction — the money in hand at that specific year. They include the effects of inflation and reflect the price levels prevailing in each future year.

When we say 'the project will generate revenues of Rs. 80,000 in year 3,' we mean Rs. 80,000 in year-3 money — which includes inflation that has occurred up to that point. These are actual cash flows as they would appear in the firm's accounts.

Constant-Worth Dollars (Real / Base-Year Dollars)

Constant-worth dollars (also called real dollars, base-year dollars, or today's dollars) represent purchasing power relative to a fixed reference year (usually year 0 or the present). They are actual dollars that have been 'deflated' to remove the effect of inflation, so all values reflect the same purchasing power as the base year.

A constant-worth dollar amount indicates how much purchasing power a future cash flow represents in terms of today's money, allowing valid comparisons across different years.

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Constant-Worth Amount in year n  =  Actual Amount in year n  /
                                    (1+f)^n
Actual Amount in year n  =  Constant-Worth Amount  ×  (1+f)^n
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Key Distinction

Basis	Actual (Nominal) Dollars	Constant-Worth (Real) Dollars
Also called	Then-current, nominal dollars	Base-year, real, today's dollars
Includes inflation?	Yes — reflects actual prices each year	No — all adjusted to base year prices
Use in analysis	When interest rate used is the market (nominal) rate	When interest rate used is the real rate
Example	Rs. 1,20,000 payable in year 5	Rs. 81,623 in year-0 purchasing power (if f=8%)
Cash flow profile	Increasing over time (due to inflation)	Constant or declining in real terms

Importance of This Distinction

The two types of dollars must be handled consistently: actual dollars must be discounted using the nominal (market) interest rate, and constant-worth dollars must be discounted using the real interest rate. Mixing the two — for example, using actual dollars with the real interest rate — produces incorrect and misleading results.

Rule: Actual dollars ↔ Nominal interest rate. Constant-worth dollars ↔ Real interest rate.

Q4. Derive the relationship between nominal (market) interest rate, real interest rate, and inflation rate. [2081 Ashwin Q11b, 2079 Chaitra Q11a, 2076 Baisakh Q11a — 4 Marks]

Answer:

Derivation

Consider Rs. 1 invested for one year. Let:

- i = nominal (market) interest rate per year (what banks advertise)
- i' = real interest rate per year (inflation-free rate of return)
- f = general inflation rate per year

After one year, Rs. 1 grows to Rs. (1 + i) in actual (nominal) dollars.

To find the real worth of this amount, we deflate back to base-year purchasing power:

$$\text{Real worth} = (1 + i) / (1 + f)$$

Since the real worth must equal 1 + i' (the real rate of return on the original Rs. 1):

$$1 + i' = (1 + i) / (1 + f)$$

The Exact Inflation-Interest Rate Relationship

$$(1 + i) = (1 + i') \times (1 + f)$$

$$\text{Expanding:} \quad i = i' + f + i' \times f$$

$$\text{Real interest rate:} \quad i' = (i - f) / (1 + f)$$

$$\text{Nominal interest rate:} \quad i = i' + f + i' \times f$$

Approximate Relationship (for small rates)

When both i' and f are small, the cross-product term i'×f is negligible, giving the approximate relationship:

$$i \approx i' + f \quad \text{(approximate, valid when } i' \text{ and } f \text{ are both small, say} < 10\%)$$

$$i' \approx i - f \quad \text{(approximate real rate)}$$

Example: Nominal rate i = 15%, inflation f = 8%.

$$\text{Exact:} \quad i' = (0.15 - 0.08)/(1 + 0.08) = 0.07/1.08 = 6.48\%$$

$$\text{Approx:} \quad i' \approx 15\% - 8\% = 7\% \quad \text{(slight overestimate)}$$

Term	Symbol	Meaning	Relationship
Nominal (market) rate	i	Observed bank/market rate — includes inflation	i = i' + f + i'f
Real interest rate	i'	Inflation-free rate of return on purchasing power	i' = (i−f)/(1+f)
Inflation rate	f	Annual rate of price level increase	f = (i−i')/(1+i')

Section B: Economic Analysis Using Constant-Worth (Real) Dollars

Q5. Explain the constant-worth dollar (real dollar) analysis method. A project costs Rs. 2,00,000 today. Annual constant-worth benefit = Rs. 60,000. Life n = 6 years; S = 0. Real MARR = 8%. Find NPW. [2082 Kartik Q12a, 2080 Ashwin Q11b — 5 Marks]

Answer:

Constant-Worth Dollar Analysis — Method

In the constant-worth (real dollar) approach, all future cash flows are expressed in terms of today's purchasing power (base-year dollars). Inflation effects are removed from the cash flows. The appropriate discount rate is the real interest rate i' (not the nominal rate i).

When cash flows are already stated in constant-worth dollars (i.e., they are assumed to maintain the same purchasing power each year regardless of inflation), the analysis is exactly the same as a standard time-value analysis — but using the real rate i' as MARR.

Numerical Solution

P = Rs. 2,00,000; Annual constant-worth benefit = Rs. 60,000/yr; n = 6 years; S = 0; Real MARR (i') = 8%

$$NPW = -P + A \times (P/A, i', n)$$

$$(P/A, 8\%, 6) = 4.6229$$

$$NPW = -2,00,000 + 60,000 \times 4.6229$$

$$= -2,00,000 + 2,77,374 = Rs.\ 77,374$$

NPW = Rs. 77,374 > 0. The project is economically acceptable at a real MARR of 8%.

Since the benefits are stated in constant-worth dollars, using the real rate ensures we are comparing like with like — real cash flows discounted at the real rate of return. This gives the true present worth in today's purchasing power.

Q6. A uniform real (constant-worth) series of Rs. 40,000 per year is expected for 8 years starting from year 1. Inflation rate f = 6%, nominal MARR = 14%. Find the NPW of this series using (a) constant-worth analysis and (b) actual-dollar analysis. Verify both give the same result. [2081 Chaitra Q11b, 2079 Jestha Q11b — 6 Marks]

Answer:

Constant-worth benefit: A' = Rs. 40,000/yr; n = 8 years; f = 6%; Nominal MARR: i = 14%

$$\text{Real interest rate:} \quad i' = (i - f)/(1 + f) = (0.14 - 0.06)/1.06 = 0.08/1.06 = 7.547\%$$

(a) Constant-Worth Analysis (using real rate i')

$$NPW = 40,000 \times (P/A, 7.547\%, 8)$$

$$(P/A, 7.547\%, 8) = [1 - (1+0.07547)^{-8}] / 0.07547$$

$$(1.07547)^8 = 1.7835; \quad (P/A) = (1 - 1/1.7835)/0.07547 = (1-0.5607)/0.07547 = 0.4393/0.07547 = 5.821$$

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NPW = 40,000 × 5.821 = Rs. 2,32,840
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(b) Actual-Dollar Analysis (using nominal rate i = 14%)

Convert constant-worth series to actual dollars: Actual amount in year k = A' × (1+f)^k

Year k	Constant-Worth A' (Rs.)	(1+f)^k = (1.06)^k	Actual Amount (Rs.)	(P/F,14%,k)	PW (Rs.)
1	40,000	1.0600	42,400	0.8772	37,193
2	40,000	1.1236	44,944	0.7695	34,584
3	40,000	1.1910	47,640	0.6750	32,157
4	40,000	1.2625	50,500	0.5921	29,901
5	40,000	1.3382	53,528	0.5194	27,802
6	40,000	1.4185	56,740	0.4556	25,850
7	40,000	1.5036	60,144	0.3996	24,033
8	40,000	1.5938	63,752	0.3506	22,352

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NPW = 37,193+34,584+32,157+29,901+27,802+25,850+24,033+22,352 =
Rs. 2,33,872
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Minor difference (Rs. 2,33,872 vs. Rs. 2,32,840) due to rounding in interest factor calculation.

Both methods give approximately the same NPW (≈ Rs. 2,33,000), confirming that: Constant-worth analysis using real rate i' ≡ Actual-dollar analysis using nominal rate i, provided they are applied consistently.

Q7. Find the real interest rate and convert between actual and constant-worth dollars. Given: nominal MARR = 18%; inflation rate f = 10%. A project yields actual cash flows of Rs. 55,000; Rs. 60,000; Rs. 70,000 in years 1, 2, 3. Find the constant-worth equivalents and the NPW using both approaches. [2081 Ashwin Q12a, 2080 Chaitra Q11b — 5 Marks]

Answer:

Nominal rate i = 18%; Inflation rate f = 10%; n = 3 years

Step 1: Find Real Interest Rate

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i' = (i − f) / (1 + f) = (0.18 − 0.10) / 1.10 = 0.08/1.10 =
7.273%
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Step 2: Convert Actual Dollars to Constant-Worth (Base Year 0) Dollars

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Constant-Worth Amount in year k  =  Actual Amount_k / (1+f)^k
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Year k	Actual Cash Flow (Rs.)	Deflation Factor 1/(1.10)^k	Constant-Worth Amount (Rs.)
1	55,000	1/1.100 = 0.9091	50,000
2	60,000	1/1.210 = 0.8264	49,587
3	70,000	1/1.331 = 0.7513	52,591

Method 1: NPW Using Actual Dollars at Nominal Rate i = 18%

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(P/F,18%,k): k=1→0.8475, k=2→0.7182, k=3→0.6086
NPW = 55,000(0.8475) + 60,000(0.7182) + 70,000(0.6086)
    = 46,613 + 43,092 + 42,602 = Rs. 1,32,307
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Method 2: NPW Using Constant-Worth Dollars at Real Rate i' = 7.273%

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(P/F, 7.273%, k): k=1→1/1.07273=0.9323, k=2→0.8692, k=3→0.8101
NPW = 50,000(0.9323) + 49,587(0.8692) + 52,591(0.8101)
    = 46,615 + 43,102 + 42,603 = Rs. 1,32,320
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Both methods give approximately the same NPW (≈ Rs. 1,32,300), confirming the equivalence of the two approaches when applied correctly.

Section C: Economic Analysis Using Actual (Nominal) Dollars

Q8. A project has a first cost of Rs. 3,00,000. Constant-worth (today's Rs.) annual benefit is Rs. 90,000/year. Life = 8 years; S = 0. Nominal MARR = 15%; Inflation = 6%. Find NPW using actual-dollar analysis. [2082 Kartik Q12b, 2079 Chaitra Q11b — 6 Marks]

Answer:

P = Rs. 3,00,000; Constant-worth benefit A' = Rs. 90,000/yr; n = 8 years; S = 0; i = 15%; f = 6%

Step 1: Convert Constant-Worth Benefits to Actual Dollars

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Actual benefit in year k  =  A' × (1+f)^k  =  90,000 × (1.06)^k
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Step 2: Calculate PW of each actual cash flow at nominal rate i = 15%

Year k	(1.06)^k	Actual Benefit (Rs.)	(P/F,15%,k)	PW of Benefit (Rs.)
1	1.0600	95,400	0.8696	82,967
2	1.1236	1,01,124	0.7561	76,460
3	1.1910	1,07,190	0.6575	70,488
4	1.2625	1,13,625	0.5718	64,990
5	1.3382	1,20,438	0.4972	59,882
6	1.4185	1,27,665	0.4323	55,173
7	1.5036	1,35,324	0.3759	50,879
8	1.5938	1,43,442	0.3269	46,886

Total PW of Benefits = 82,967+76,460+70,488+64,990+59,882+55,173+50,879+46,886 = Rs. 5,07,725

Step 3: NPW

NPW = −3,00,000 + 5,07,725 = Rs. 2,07,725

NPW = Rs. 2,07,725 > 0. The project is economically acceptable.

Verification Using Real Rate

i' = (0.15 − 0.06)/1.06 = 8.491%

(P/A, 8.491%, 8) = [1−(1.08491)^{−8}]/0.08491

(1.08491)^8 = 1.9011; (P/A) = (1−0.5260)/0.08491 = 5.582

NPW = −3,00,000 + 90,000(5.582) = −3,00,000+5,02,380 = Rs. 2,02,380

Close to the Rs. 2,07,725 (small difference due to rounding). Both methods confirm strong project viability.

Q9. A machine costs Rs. 1,50,000 today. In actual (then-current) dollars, annual revenues are: yr1=Rs. 50,000; yr2=Rs. 55,000; yr3=Rs. 60,000; yr4=Rs. 65,000; yr5=Rs. 70,000. Annual O&M = Rs. 12,000 in year-0 dollars (growing with inflation). f = 7%, nominal MARR = 15%, n = 5 years, S = 0. Find NPW. [2081 Chaitra Q12, 2079 Jestha Q12 — 6 Marks]

Answer:

P = Rs. 1,50,000; f = 7%; i = 15%; n = 5 years; S = 0

Revenues already given in actual dollars. O&M in year-0 dollars = Rs. 12,000/yr → convert to actual:

Convert O&M to Actual Dollars

Year k	O&M (year-0 Rs.)	(1.07)^k	Actual O&M (Rs.)
1	12,000	1.0700	12,840
2	12,000	1.1449	13,739
3	12,000	1.2250	14,700
4	12,000	1.3108	15,730
5	12,000	1.4026	16,831

Net Actual BTCF (Revenue − O&M) and NPW

Year	Revenue (Rs.)	Actual O&M (Rs.)	Net ATCF (Rs.)	(P/F,15%,k)	PW (Rs.)
1	50,000	12,840	37,160	0.8696	32,313
2	55,000	13,739	41,261	0.7561	31,198
3	60,000	14,700	45,300	0.6575	29,785
4	65,000	15,730	49,270	0.5718	28,176
5	70,000	16,831	53,169	0.4972	26,435

Total PW of Net Cash Flows = 32,313+31,198+29,785+28,176+26,435 = Rs. 1,47,907

NPW = −1,50,000 + 1,47,907 = −Rs. 2,093

NPW = −Rs. 2,093 < 0. The project is marginally unacceptable at 15% nominal MARR. The project fails to meet MARR when both inflation-adjusted costs and time value are considered.

Q10. A project's constant-worth revenues start at Rs. 80,000/yr and are expected to remain constant (purchasing power maintained). Inflation = 8%. Nominal MARR = 16%. First cost = Rs. 4,00,000; n = 10 years; S = Rs. 50,000. Find (a) actual cash flows, (b) NPW. [2080 Ashwin Q12, 2078 Baishakh Q11 — 6 Marks]

Answer:

P = Rs. 4,00,000; Constant-worth A' = Rs. 80,000/yr; f = 8%; i = 16%; n = 10 years; S in year-0 dollars = Rs. 50,000

(a) Actual Revenues in Year k

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Actual Revenue_k = 80,000 × (1.08)^k
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Actual Salvage in year 10 = 50,000 × (1.08)^10 = 50,000 × 2.1589 = Rs. 1,07,945

(b) NPW using actual dollars at i = 16%

Rather than compute year by year, use the combined factor for a constant-worth series inflating at f, discounted at i:

```
For a constant-worth annuity growing with inflation:
PW = A' × (P/A, i', n)  where i' = (i−f)/(1+f) =
(0.16−0.08)/1.08 = 7.407%
(P/A, 7.407%, 10) = [1−(1.07407)^{−10}]/0.07407
(1.07407)^{10} = 2.0394;   (P/A) = (1−0.4903)/0.07407 =
0.5097/0.07407 = 6.881
PW of revenues = 80,000 × 6.881 = Rs. 5,50,480
PW of actual salvage: (P/F,16%,10) = 1/(1.16)^10 = 1/4.4114 =
0.2267
PW of salvage = 50,000 × (1.08)^10 × (P/F,16%,10) = 1,07,945 ×
0.2267 = Rs. 24,471
NPW = −4,00,000 + 5,50,480 + 24,471 = Rs. 1,74,951
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NPW = Rs. 1,74,951 > 0. The project is economically acceptable at 16% nominal MARR.

Section D: Annual Worth Analysis with Inflation

Q11. Find the annual worth (AW) of a project under inflation. P = Rs. 5,00,000; S = Rs. 60,000 (year-0 Rs.); n = 10 years; Annual benefit = Rs. 1,20,000 (constant-worth); f = 7%; Nominal MARR = 15%. [2082 Kartik Q12c, 2080 Chaitra Q12, 2079 Ashwin Q12 — 6 Marks]

Answer:

P = Rs. 5,00,000; A' = Rs. 1,20,000/yr (constant-worth); n = 10 years; f = 7%; i = 15%

Actual salvage in year 10 = 60,000 × (1.07)^10 = 60,000 × 1.9672 = Rs. 1,18,032

Method: Compute NPW first, then convert to AW

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i' = (0.15 − 0.07)/1.07 = 0.08/1.07 = 7.477%
```

$$(P/A,\ 7.477\%,\ 10):\ (1.07477)^{10} = 2.0603;\quad (P/A) = (1-0.4854)/0.07477 = 6.882$$

$$\text{PW of constant-worth benefits} = 1{,}20{,}000 \times 6.882 = \text{Rs. } 8{,}25{,}840$$

$$\text{PW of actual salvage} = 1{,}18{,}032 \times (P/F,15\%,10)$$

$$(P/F,15\%,10) = 1/(1.15)^{10} = 1/4.0456 = 0.2472$$

$$\text{PW of salvage} = 1{,}18{,}032 \times 0.2472 = \text{Rs. } 29{,}178$$

$$NPW = -5{,}00{,}000 + 8{,}25{,}840 + 29{,}178 = \text{Rs. } 3{,}55{,}018$$

Convert NPW to AW (Nominal AW at nominal rate)

$$(A/P,\ 15\%,\ 10) = 0.19925$$

$$\text{Nominal AW} = NPW \times (A/P,\ 15\%,\ 10) = 3{,}55{,}018 \times 0.19925 = \text{Rs. } 70{,}762/\text{year}$$

This AW of Rs. 70,762 is in nominal (actual) dollars — it is the equivalent annual amount the project generates in future-year money, accounting for inflation.

Real (Constant-Worth) AW

$$\text{Real AW} = NPW \times (A/P,\ i',\ 10) = 3{,}55{,}018 \times (A/P,\ 7.477\%,\ 10)$$

$$(A/P,\ 7.477\%,\ 10) = 0.07477 \times 2.0603/(2.0603-1) = 0.15407/1.0603 = 0.14531$$

$$\text{Real AW} = 3{,}55{,}018 \times 0.14531 = \text{Rs. } 51{,}589/\text{year}$$

Nominal AW = Rs. 70,762/year (in each year's actual dollars). Real AW = Rs. 51,589/year (in constant year-0 purchasing power). Both indicate a profitable project.

Q12. Explain capital recovery with inflation. A machine costs Rs. 2,00,000 with salvage Rs. 20,000 after 8 years. Inflation = 6%, nominal MARR = 14%. Find the annual capital recovery (CR) amount in (a) nominal dollars, (b) constant-worth dollars. [2081 Ashwin Q12b, 2078 Chaitra Q11b — 5 Marks]

Answer:

Capital Recovery with Inflation — Concept

Capital recovery (CR) is the equivalent annual amount that must be earned to recover the initial investment and any opportunity cost (interest on invested capital) over the asset's life. When inflation is present, we must decide whether to compute CR in nominal (actual) dollars or in constant-worth (real) dollars, and use the corresponding interest rate.

P = Rs. 2,00,000; S_0 = Rs. 20,000 (in today's Rs.); n = 8 years; f = 6%; i = 14%

$$i' = (0.14 - 0.06)/1.06 = 7.547\%$$

Actual salvage in year 8 = 20,000 × (1.06)^8 = 20,000 × 1.5938 = Rs. 31,876

(a) Nominal Capital Recovery (CR in actual dollars)

```
CR_nominal = P(A/P, i, n) - S_actual(A/F, i, n)
(A/P, 14%, 8) = 0.21557;   (A/F, 14%, 8) = 0.07557
CR_nominal = 2,00,000(0.21557) - 31,876(0.07557)
= 43,114 - 2,409 = Rs. 40,705/year (nominal dollars)
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(b) Real (Constant-Worth) Capital Recovery

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CR_real = P(A/P, i', n) - S_0(A/F, i', n)
(A/P, 7.547%, 8): (1.07547)^8=1.7835;
(A/P)=0.07547×1.7835/(1.7835-1)=0.1346/0.7835=0.17180
(A/F, 7.547%, 8): (A/F)=(A/P)-i'=0.17180-0.07547=0.09633
CR_real = 2,00,000(0.17180) - 20,000(0.09633)
= 34,360 - 1,927 = Rs. 32,433/year (constant-worth
dollars)
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Nominal CR = Rs. 40,705/year (in each year's actual dollars — includes inflation). Real CR = Rs. 32,433/year (in constant year-0 purchasing power). The nominal CR is higher because it includes inflationary growth.

Section E: Effect of Inflation on After-Tax Economic Analysis

Q13. Explain how inflation affects after-tax economic analysis. A project has BTCF = Rs. 60,000/year (constant-worth). B = Rs. 2,00,000; S = 0; n = 5 years; t = 35%; f = 8%; Nominal MARR = 16%. SL depreciation. Compute AT-NPW. [2080 Chaitra Q12b, 2079 Chaitra Q12, 2077 Chaitra Q12 — 8 Marks]

Answer:

How Inflation Affects After-Tax Analysis

Inflation has a particularly insidious effect on after-tax project economics because depreciation charges are based on the original (historical) cost in nominal dollars — they do not increase with inflation. This means:

1. Taxable income is overstated: As inflation pushes actual revenues higher, depreciation (fixed in nominal terms) provides a diminishing real tax shield over time.

2. Real after-tax cash flows erode: The real (purchasing power) value of after-tax cash flows is reduced more under inflation than a simple before-tax analysis would suggest.
3. The purchasing power of depreciation tax shields falls: Even though the nominal depreciation amount is fixed, its real value (purchasing power) declines with inflation — reducing the real tax benefit.

This is known as the 'inflation tax penalty' — inflation effectively increases the real tax burden on capital investments. Projects that appear acceptable before considering inflation may become marginal or unacceptable when inflation is properly included.

Numerical Solution

B = Rs. 2,00,000; S = 0; n = 5 years; BTCF (constant-worth) = Rs. 60,000/yr; t = 35%; f = 8%; i = 16%

SL Depreciation: d = 2,00,000/5 = Rs. 40,000/year (fixed nominal, does NOT increase with inflation)

Convert constant-worth BTCF to actual dollars: Actual BTCF in year k = 60,000 × (1.08)^k

Year-by-Year ATCF Calculation

Year k	Actual BTCF (Rs.)	d (Rs.)	TI = BTCF−d (Rs.)	Tax @35% (Rs.)	Actual ATCF (Rs.)	(P/F,16%,k)	PW (Rs.)
1	64,800	40,000	24,800	8,680	56,120	0.8621	48,381
2	69,984	40,000	29,984	10,494	59,490	0.7432	44,214
3	75,583	40,000	35,583	12,454	63,129	0.6407	40,461
4	81,629	40,000	41,629	14,570	67,059	0.5523	37,035
5	88,159	40,000	48,159	16,856	71,303	0.4761	33,957

Year 1: Actual BTCF = 60,000 × 1.08 = 64,800. TI = 64,800 − 40,000 = 24,800. Tax = 8,680. ATCF = 56,120.

Total PW of ATCFs = 48,381+44,214+40,461+37,035+33,957 = Rs. 2,04,048

AT-NPW = −2,00,000 + 2,04,048 = Rs. 4,048

AT-NPW = Rs. 4,048 > 0 (marginally acceptable).

Comparison: Without vs. With Inflation

Without inflation (f=0): Actual BTCF = constant Rs. 60,000. i'=(16−0)/1=16%. AT-NPW would be much higher.

With inflation (f=8%): AT-NPW falls sharply to only Rs. 4,048 — because the depreciation tax shield does not grow with inflation, creating a significant inflation tax penalty. This demonstrates that inflation can transform a clearly profitable project into a barely acceptable one.

Q14. A project with B = Rs. 1,50,000 (constant-worth benefit Rs. 55,000/yr) shows AT-NPW = Rs. 28,500 without inflation. If inflation = 9% and nominal MARR = 18%, t = 40%, SL depreciation, S = 0, n = 5 years, recalculate AT-NPW and show the inflation penalty. [2081 Chaitra Q12b, 2079 Ashwin Q12 — 6 Marks]

Answer:

B = Rs. 1,50,000; A' = Rs. 55,000/yr (constant-worth); S = 0; n = 5 yrs; t = 40%; f = 9%; i = 18%

d_SL = 1,50,000/5 = Rs. 30,000/year (fixed in nominal dollars)

Actual BTCF in year k = 55,000 × (1.09)^k

Year k	(1.09)^k	Actual BTCF (Rs.)	d (Rs.)	TI (Rs.)	Tax @40% (Rs.)	Actual ATCF (Rs.)
1	1.0900	59,950	30,000	29,950	11,980	47,970
2	1.1881	65,346	30,000	35,346	14,138	51,208
3	1.2950	71,225	30,000	41,225	16,490	54,735
4	1.4116	77,638	30,000	47,638	19,055	58,583
5	1.5386	84,623	30,000	54,623	21,849	62,774

(P/F,18%,k): k=1→0.8475, k=2→0.7182, k=3→0.6086, k=4→0.5158, k=5→0.4371

AT-NPW = −1,50,000 + 47,970(0.8475)+51,208(0.7182)+54,735(0.6086)+58,583(0.5158)+62,774(0.4371)

= −1,50,000 + 40,654+36,778+33,311+30,218+27,439 = Rs. 18,400

Scenario	AT-NPW (Rs.)	Difference
Without inflation (f=0)	28,500 (given)	Base
With inflation (f=9%)	18,400	−Rs. 10,100 (inflation penalty)

Inflation reduces AT-NPW by Rs. 10,100 (35% reduction). The fixed depreciation tax shield loses purchasing power under inflation, creating a significant inflation tax penalty. Projects should always be evaluated with realistic inflation assumptions.

Section F: Comprehensive Inflation Analysis Problems

Q15. A construction firm is evaluating a project. First cost = Rs. 8,00,000. Constant-worth annual benefit = Rs. 2,20,000. Annual O&M (year-0 Rs.) = Rs. 40,000. Salvage (year-0 Rs.) = Rs. 80,000. n = 10 years. f = 7%; Nominal MARR = 15%. Find NPW and AW using the actual-dollar approach. [2082 Kartik Q13, 2081 Ashwin Q13 — 8 Marks]

Answer:

P = Rs. 8,00,000; A' = Rs. 2,20,000/yr (constant-worth benefit); O&M_0 = Rs. 40,000/yr; S_0 = Rs. 80,000; n = 10 years; f = 7%; i = 15%

Step 1: Real Interest Rate

```
i' = (0.15 − 0.07)/1.07 = 7.477%
```

Step 2: PW of Benefits (Constant-Worth Annuity, using real rate)

```
(P/A, 7.477%, 10): (1.07477)^{10}=2.0603;
(P/A)=(1−0.4854)/0.07477=6.882

PW_benefit = 2,20,000 × 6.882 = Rs. 15,14,040
```

Step 3: PW of O&M Costs (also constant-worth, rising with inflation)

```
O&M grows at f per year → constant-worth annuity → use real rate

PW_O&M = 40,000 × (P/A, 7.477%, 10) = 40,000 × 6.882 = Rs.
2,75,280
```

Step 4: PW of Salvage (year-0 Rs. 80,000 growing with inflation)

```
Actual salvage in year 10 = 80,000 × (1.07)^{10} = 80,000 ×
1.9672 = Rs. 1,57,376

(P/F, 15%, 10) = 0.2472

PW_salvage = 1,57,376 × 0.2472 = Rs. 38,904
```

Step 5: NPW

```
NPW = −8,00,000 + 15,14,040 − 2,75,280 + 38,904
    = −8,00,000 + 12,77,664 = Rs. 4,77,664
```

Step 6: AW (Nominal and Real)

```
Nominal AW = NPW × (A/P, 15%, 10) = 4,77,664 × 0.19925 = Rs.
95,174/year
```

Real AW = NPW × (A/P, 7.477%, 10) = 4,77,664 × 0.14531 = Rs. 69,408/year

NPW = Rs. 4,77,664 > 0. Nominal AW = Rs. 95,174/yr. Real AW = Rs. 69,408/yr. The project is strongly acceptable.

Q16. Compare two alternatives under inflation using AW method. Nominal MARR = 12%; f = 6%. [2080 Ashwin Q13, 2078 Chaitra Q12 — 8 Marks]

Answer:

	Alt. X	Alt. Y
First Cost (Rs.)	1,50,000	2,80,000
Annual Benefit (constant-worth Rs.)	48,000	80,000
Annual O&M (constant-worth Rs.)	10,000	16,000
Salvage (year-0 Rs.)	15,000	30,000
Life (years)	6	10

Nominal MARR = 12%; f = 6%; Unequal lives — use AW over each alternative's own life (with repeatability assumption).

i' = (0.12 − 0.06)/1.06 = 5.660%

Annual Worth of Alt. X (n=6 years)

Net constant-worth benefit = 48,000 − 10,000 = Rs. 38,000/yr

Actual salvage = 15,000 × (1.06)^6 = 15,000 × 1.4185 = Rs. 21,278

(P/A, 5.660%, 6): (1.05660)^6=1.3929; (P/A)=(1−0.7179)/0.05660=4.980

PW_net_benefit_X = 38,000 × 4.980 = Rs. 1,89,240

(P/F,12%,6)=0.5066; PW_salvage_X = 21,278 × 0.5066 = Rs. 10,779

NPW_X = −1,50,000 + 1,89,240 + 10,779 = Rs. 50,019

AW_X = NPW_X × (A/P,5.660%,6) = 50,019 × (1/4.980) = 50,019 × 0.20080 = Rs. 10,044/yr (real)

Annual Worth of Alt. Y (n=10 years)

Net constant-worth benefit = 80,000 − 16,000 = Rs. 64,000/yr

Actual salvage = 30,000 × (1.06)^{10} = 30,000 × 1.7908 = Rs. 53,724

```
(P/A, 5.660%, 10): (1.05660)^{10}=1.7339;
(P/A)=(1-0.5769)/0.05660=7.475
PW_net_benefit_Y = 64,000 × 7.475 = Rs. 4,78,400
(P/F,12%,10)=0.3220;  PW_salvage_Y = 53,724 × 0.3220 = Rs.
17,299
NPW_Y = -2,80,000 + 4,78,400 + 17,299 = Rs. 2,15,699
AW_Y = NPW_Y × (A/P,5.660%,10) = 2,15,699 × (1/7.475) = 2,15,699
× 0.13378 = Rs. 28,857/yr (real)
```

Alternative	NPW (Rs.)	Real AW (Rs./yr)	Decision
Alt. X	50,019	10,044	Acceptable
Alt. Y	2,15,699	28,857	Best — select Y

Select Alternative Y. Its real AW of Rs. 28,857/yr is nearly three times that of Alt. X, indicating substantially better long-term economic performance even after accounting for the higher initial investment.

Section G: Conceptual and Theory Questions

Q17. What is the 'inflation tax penalty'? Illustrate why inflation reduces the real after-tax value of depreciation tax shields. [2079 Jestha Q11c, 2076 Baisakh Q12 — 3 Marks]

Answer:

The Inflation Tax Penalty — Concept

The inflation tax penalty is the reduction in real (purchasing power) after-tax returns caused by inflation's interaction with depreciation-based tax deductions. It arises because depreciation is calculated on the historical (nominal) cost of an asset — it does not increase with inflation — while actual revenues and costs grow with inflation. This creates an asymmetry: revenues grow, but the tax shield from depreciation stays fixed in nominal terms, meaning its real value declines each year.

Illustration

Suppose a machine costs Rs. 1,00,000 and is depreciated straight-line over 5 years: d = Rs. 20,000/year. The nominal depreciation tax shield at t = 40% is:

```
Nominal tax shield per year = 0.40 × 20,000 = Rs. 8,000/year
(fixed)
```

The real (purchasing power) value of this Rs. 8,000 shield decreases each year with inflation at 8%:

Year	Nominal Tax Shield (Rs.)	Deflation Factor 1/(1.08)^k	Real Value of Shield (Rs.)
1	8,000	0.9259	7,407
2	8,000	0.8573	6,858
3	8,000	0.7938	6,350
4	8,000	0.7350	5,880
5	8,000	0.6806	5,445

The total nominal tax shield = Rs. 40,000. But its total real value = 7,407+6,858+6,350+5,880+5,445 = Rs. 31,940. Inflation has eroded Rs. 8,060 (20%) of the tax shield's real value.

This is the inflation tax penalty: the government effectively captures a larger share of the real return as inflation rises, because it taxes nominal income but the depreciation deduction is fixed in nominal terms.

Q18. What is meant by 'consistent analysis' in the presence of inflation? What errors arise when actual dollars are mixed with the real interest rate (or vice versa)? [2078 Chaitra Q11c, 2075 Bhadra Q11 — 3 Marks]

Answer:

Consistent Analysis — Meaning

Consistent analysis in an inflationary environment means always pairing the correct type of cash flow with the correct type of interest rate:

Cash Flow Type	Correct Interest Rate	Reason
Actual (Nominal/Then-Current) Dollars	Nominal (Market) Rate i	Both include inflation effects — they offset correctly
Constant-Worth (Real/Base-Year) Dollars	Real Rate i'	Both exclude inflation effects — they match correctly

Errors from Inconsistent Analysis

- Using actual dollars with the real interest rate: This double-counts inflation — the cash flows include inflation, but the discount rate does not account for it. The result is a grossly overestimated NPW (artificially high). Projects appear far more attractive than they truly are.

- Using constant-worth dollars with the nominal interest rate: This penalises the cash flows twice — they are already deflated, but now discounted at a higher rate that includes inflation. The result is an underestimated NPW. Projects appear worse than they truly are.

Example: If actual revenue is Rs. 1,08,000 in year 1 (= Rs. 1,00,000 × 1.08 for f=8%) and we (wrongly) discount at the real rate i'=7%: PW = 1,08,000/1.07 = Rs. 1,00,935. Correct: PW = 1,08,000/1.15 = Rs. 93,913 (using nominal i=15%). The error: Rs. 1,00,935 − Rs. 93,913 = Rs. 7,022 overestimation per Rs. 1,00,000 of constant revenue.

Rule to memorise: Actual $ ↔ Nominal rate i. Real $ ↔ Real rate i'. Never mix these pairs.

Quick Reference — Chapter 8 Key Formulas

Formula	Expression	Application
Inflation rate	f_n = (CPI_n − CPI_{n-1})/CPI_{n-1}	Annual inflation from CPI data
Nominal–Real–Inflation	(1+i) = (1+i')(1+f)	Convert between i, i', and f
Real rate (exact)	i' = (i−f)/(1+f)	Find inflation-free rate
Real rate (approx.)	i' ≈ i − f	Valid when i,f both small
Actual to CW	CW_k = Actual_k / (1+f)^k	Deflate actual dollars to base year
CW to Actual	Actual_k = CW × (1+f)^k	Inflate base-year dollars to year k
Purchasing power	PP = Amount / (1+f)^n	Real value n years from now
CW annuity NPW	A' × (P/A, i', n)	For constant-worth annual series
Nominal CR	P(A/P,i,n) − S_act(A/F,i,n)	Capital recovery in actual $
Real CR	P(A/P,i',n) − S_0(A/F,i',n)	Capital recovery in CW $

- Actual dollars ↔ Nominal interest rate i. Constant-worth dollars ↔ Real interest rate i'.
- Both consistent approaches give the same NPW — never mix the two.
- Inflation erodes the real value of fixed depreciation tax shields — creating an inflation tax penalty.
- Under inflation, after-tax project returns are lower than before-tax analysis suggests, because depreciation does not inflate but taxable revenues do.
- Higher inflation rates generally reduce AT-NPW for capital-intensive projects with long depreciation periods.

—— End of Chapter 8: Inflation and Its Effects on Engineering Economy ——